

LEARNING PLAN

Unit Title: DIGITAL ELECTRONICS I	Unit code: ENG/CU/EI/CC/05/5/MA
Name of trainer: MARGARET	
Institution: ABERDARE TECHNICAL COLLEGE	Level: 5,6
Date of preparation: 28/8/2026	Date of revision:
Number of learners:	Class: MODULE IV
Skill or job task: 1. Apply knowledge of number systems 2. Apply knowledge of binary codes 3. Apply logic gates and Boolean algebra concepts	
Bench marks or performance criteria: 1. APPLY KNOWLEDGE OF NUMBER SYSTEMS 1.1 Number systems 1.1.1 Decimal number system 1.1.2 Octal number system 1.1.3 Hexadecimal number system 1.1.4 Binary number system 1.2 Number system representation 1.3 Conversion of number systems	

1.4 Perform 1's complement

1.5 Perform 2's complement

2. APPLY KNOWLEDGE OF BINARY CODES

2.1 Binary Coded Decimal (BCD)

2.2 Gray Code

2.3 Excess-3 Code

3. APPLY LOGIC GATES AND BOOLEAN ALGEBRA CONCEPTS

3.1 Basic logic gates

3.2 Universal logic gates operation

3.3 Special purpose gates

3.4 Laws of Boolean algebra

3.5 Logic expressions

3.6 Logic expression simplification

3.7 Karnaugh Maps (K-MAPS)

Week	Session No.	Session Title	Specific Learning outcomes	Key Learning Points	Trainee Activities	Learning Resources	Learning checks/ assessment	Reflection's & Date
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1	1–2	Introduction to Number Systems	By the end of the session the trainee should be able to: 1. Define a number system. 2. Identify common number systems. 3. Explain decimal number representation. 4. Identify place values in different bases.	• Meaning of number systems • Radix/base • Positional notation • Decimal system • Place values • Applications of number systems in digital electronics	1. Participate in discussion on number systems. 2. Identify different number systems from given examples. 3. Solve decimal representation exercises. 4. Identify place values.	• Whiteboard • Marker • Digital electronics textbooks • Handouts • Calculators • Charts	• Oral questioning • Written exercises • Observation	
	3–4	Binary, Octal and Hexadecimal Number Systems	1. Explain binary, octal and hexadecimal systems. 2. Represent numbers in different bases.	• Binary system – base 2 • Octal system – base 8 • Hexadecimal system – base 16	1. Convert decimal numbers into binary, octal and hexadecimal. 2. Convert binary numbers into octal and hexadecimal.	• Whiteboard • Number system charts • Calculators • Textbooks • Handouts	• Written exercises • Oral questioning • Practical conversion exercises • Observation	

2	1–2		3. Convert numbers between different number systems. 4. Apply number system conversions in digital applications.	• Positional values • Binary-octal conversion • Binary-hexadecimal conversion • Decimal conversions	3. Complete conversion exercises individually and in groups. 4. Discuss applications of different number systems.	• Whiteboard • Number system charts • Calculators • Textbooks • Handouts	• Written exercises • Oral questioning • Practical conversion exercises • Observation	
	3–4	Number System Conversion and Complements	1. Perform number system conversions accurately. 2. Explain 1's complement.	• Conversion procedures • 1's complement • Inverting binary bits • 2's complement	1. Perform conversion exercises. 2. Calculate 1's complement of binary numbers.	• Whiteboard • Calculators • Binary charts • Exercise sheets • Digital electronics textbooks	• Written test • Practical exercises • Oral questioning • Observation	

3	1-2		3. Perform 1's complement operations. 4. Explain and perform 2's complement operations.	• Adding 1 to 1's complement • Applications of complements in digital systems	3. Calculate 2's complement. 4. Solve signed binary arithmetic exercises.	• Whiteboard • Calculators • Binary charts • Exercise sheets • Digital electronics	• Written test • Practical exercises • Oral questioning • Observation	
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	3–4	Binary Coded Decimal (BCD)	1. Define BCD. 2. Explain BCD representation. 3. Convert decimal numbers into BCD. 4. Convert BCD into decimal numbers.	• Meaning of BCD • 8421 BCD code • BCD representation • BCD encoding • BCD decoding • BCD applications	1. Represent decimal numbers using BCD. 2. Decode BCD numbers. 3. Compare BCD with ordinary binary representation. 4. Complete BCD exercises.	• BCD charts • Whiteboard • Calculators • Textbooks • Handouts	• Written exercises • Oral questioning • Demonstration • Observation	
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4	1–2	Gray Code and Excess-3 Code	1. Define Gray Code. 2. Convert binary to Gray Code. 3. Convert Gray Code to binary. 4. Explain and apply Excess-3 Code.	• Gray Code characteristics • Binary-to-Gray conversion • Gray-to-binary conversion • Excess-3 Code • Excess-3 encoding and decoding • Applications of binary codes	1. Convert binary numbers into Gray Code. 2. Convert Gray Code into binary. 3. Encode decimal numbers using Excess-3. 4. Decode Excess-3 numbers.	• Code charts • Whiteboard • Calculators • Digital electronics textbooks • Worksheets	• Practical exercises • Written test • Oral questioning • Observation	
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	3–4	FORMATIVE ASSESSMENT 1 – BINARY CODES	1. Demonstrate knowledge of binary codes. 2. Perform BCD conversions. 3. Perform Gray Code conversions. 4. Perform Excess-3 conversions accurately.	• BCD • Gray Code • Excess-3 Code • Encoding and decoding • Code conversion procedures • Applications of binary codes	1. Answer written questions. 2. Perform practical code conversion tasks. 3. Demonstrate code conversion procedures. 4. Correct errors identified during assessment.	• Question papers • Assessment rubrics • Calculators • Code charts • Worksheets	Knowledge: Written/oral test Skills: Practical test Attitudes: Observation	
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5	1–2	Basic Logic Gates	1. Identify basic logic gates. 2. Draw logic gate symbols. 3. Construct truth tables. 4. Explain the operation of AND, OR and NOT gates.	• Digital logic concept • AND gate • OR gate • NOT gate • Logic symbols • Truth tables • Boolean expressions • Gate operation	1. Draw symbols of basic gates. 2. Construct truth tables. 3. Assemble simple logic gate circuits. 4. Test gate outputs using LEDs or logic indicators.	• Digital trainer • Logic gate ICs • Breadboard • LEDs • Switches • Connecting wires • Multimeter	• Practical demonstration • Observation • Oral questioning • Written exercises	
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	3–4	Universal and Special Purpose Logic Gates	1. Explain NAND and NOR gates. 2. Construct NAND and NOR truth tables. 3. Explain XOR and XNOR gates. 4. Apply universal gates in digital circuits.	• NAND gate • NOR gate • Universal gate concept • NAND/NOR truth tables • XOR gate • XNOR gate • Special purpose gate applications	1. Construct NAND and NOR circuits. 2. Verify truth tables. 3. Construct XOR and XNOR circuits. 4. Compare outputs of different gates.	• Digital electronics trainer • Logic ICs • Breadboard • LEDs • Switches • Multimeter • Connecting wires	• Practical demonstration • Projects • Written test • Oral questioning	
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6	1–2	Boolean Algebra and Logic Expressions	1. State the laws of Boolean algebra. 2. Write Boolean expressions from logic circuits. 3. Simplify Boolean expressions. 4. Apply Boolean algebra laws to digital circuits.	• Boolean variables • AND, OR and NOT operations • Identity law • Null law • Idempotent law • Complement law • Involution law • De Morgan's laws • Logic expressions	1. Write Boolean expressions. 2. Apply Boolean algebra laws. 3. Simplify logic expressions. 4. Verify simplified expressions using truth tables.	• Whiteboard • Boolean algebra charts • Digital trainer • Logic gates • Textbooks • Worksheets	• Written exercises • Oral questioning • Practical demonstration • Observation	
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	3–4	Karnaugh Maps (K-MAPS)	1. Explain the purpose of K-Maps. 2. Construct 2, 3 and 4-variable K-Maps. 3. Group adjacent cells correctly 4. Simplify Boolean expressions using K-Maps.	• Purpose of K-Maps • K-Map structure • Minterms • Truth-table conversion • Grouping rules • Simplification • 2, 3 and 4-variable K-Maps	1. Construct K-Maps from truth tables. 2. Enter values into K-Maps. 3. Identify groups of 1s. 4. Derive simplified Boolean expressions.	• K-Map worksheets • Whiteboard • Logic trainer • Boolean algebra textbooks • Calculators	• Written exercises • Practical demonstration • Oral questioning • Observation	
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7	3–4	FORMATIVE ASSESSMENT 2 / INTEGRATED REVISION	1. Demonstrate knowledge of logic gates. 2. Apply Boolean algebra concepts. 3. Simplify logic expressions. 4. Solve K-Map problems. 5. Demonstrate practical digital logic skills.	• Number systems • Binary codes • Logic gates • Boolean algebra • Logic expression simplification • K-MAPS • Digital circuit applications	1. Answer written and oral questions. 2. Perform practical digital logic tasks. 3. Simplify Boolean expressions. 4. Solve K-Map problems. 5. Demonstrate construction and testing of a simple logic circuit.	• Assessment papers • Rubrics • Digital trainer • Logic ICs • Breadboard • Multimeter • Worksheets	Knowledge: Written/oral test Skills: Practical assessment Attitudes: Observation	
13	SUMMATIVE CDACC ASSESSMENT							

TRAINER: MD.MARGARET

APPROVED BY: _____ SIGNATURE: _____

SIGNATURE: _____ DATE: _____

DATE: _____